

Twin-Thruster Boat Dynamics and Control Feasibility

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1. Project objective

An autonomous surface vessel (ASV) must reach a target while wind changes its motion. Two independent thrusters provide surge force and yaw moment, so guidance must turn the hull to correct lateral error. This report reproduces the presentation experiments and develops a hardware architecture for BlueBoat [1], [2].

- The slide benchmark establishes the control problem.
 - Equal thrust reaches 1.345 m/s; crosswind causes 21.23 m of unpowered drift in 60 s.
 - The nominal 20 m approach accumulates 32.7 m of crosswind error at 90 s.
- BlueBoat is the deployment platform.
 - Its dimensions and propulsion differ from the slide model. Sections 2–7 retain the original benchmark; the later sections define the selected hardware and identification work.

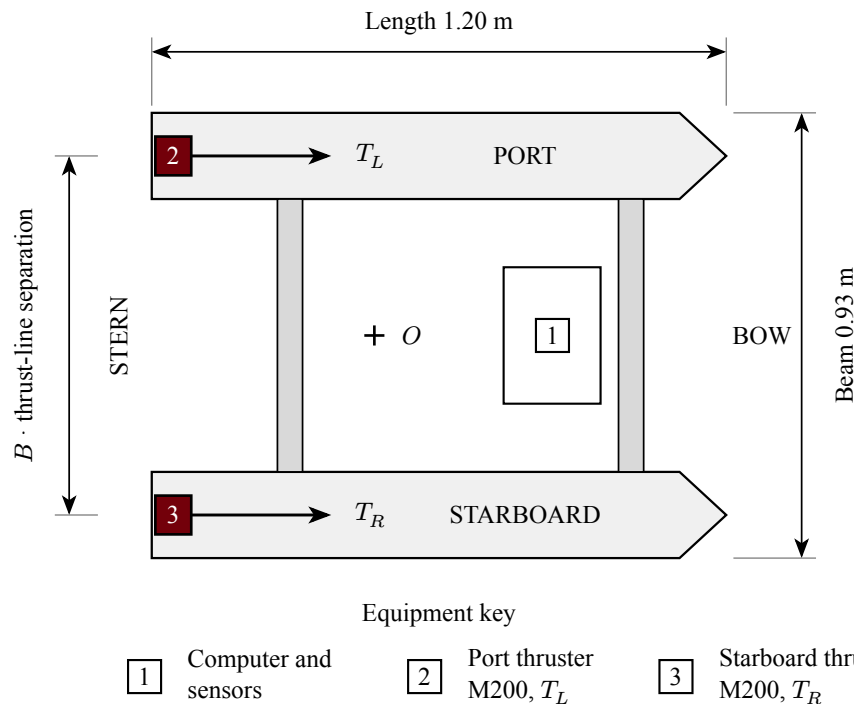


Figure 1: BlueBoat plan view with the bow to the right. The bottom key identifies the computer and sensors (1), port thruster (2), and starboard thruster (3). Both positive thrusts point forward. O denotes the hull reference origin, and B denotes thrust-line separation. The envelope uses published length and beam; internal footprints are schematic.

Quantity	Slide benchmark	BlueBoat platform
Length / beam	4.9 m / unspecified	1.20 m / 0.93 m
Mass	180 kg	14.5 kg bare; weigh loaded hull
Yaw inertia / spacing	446 kg m ² / 2.4 m	Identify I_z ; measure B
Propulsion	–100 to 250 N each	Two M200 weedless propellers

2. Planar model and simulation method

The twin-thruster configuration in Figure 1 is modeled with three degrees of freedom. This section uses the slide benchmark parameters in the preceding table. The inertial pose is $\boldsymbol{\eta} = (x, y, \psi)^T$, and the body velocity is $\boldsymbol{\nu} = (u, v, r)^T$. Here, x and y denote east and north position, ψ denotes heading measured counterclockwise from east, and u , v , and r denote surge speed, sway speed, and yaw rate. This planar reduction follows the rigid-body, Coriolis, and damping structure of the marine-craft equations [3].

The heading rotates body velocity into the inertial frame. The resulting kinematics are

$$\dot{x} = u \cos \psi - v \sin \psi, \quad \dot{y} = u \sin \psi + v \cos \psi, \quad \dot{\psi} = r. \quad (1)$$

The left and right thrusters apply T_L and T_R . Because their force lines are separated by B , the force and moment allocation is

$$\tau_u = T_L + T_R, \quad \tau_v = 0, \quad \tau_r = \frac{B}{2}(T_R - T_L). \quad (2)$$

The nonlinear equations retain the coupling generated by rotation. With body-frame wind forces F_u and F_v , the equations are

$$m\dot{u} = mvr + T_L + T_R - d_1 u - d_2 u|u| + F_u, \quad (3)$$

$$m\dot{v} = -mur - d_v v + F_v, \quad (4)$$

$$I_z \dot{r} = \frac{B}{2}(T_R - T_L) - d_r r. \quad (5)$$

Coefficient	Value	Reconstruction basis
Linear surge drag, d_1	51.3 N s/m	Stated in the slides
Quadratic surge drag, d_2	72.4 N s ² /m ²	Stated in the slides
Linear sway drag, d_v	40 N s/m	Stated in the slides
Linear yaw drag, d_r	400 N m s	Inferred from yaw step and pulse
5 m/s wind force	15.3 N	Slide-plot value

For a constant inertial wind force (F_x, F_y) , the body components are $F_u = F_x \cos \psi + F_y \sin \psi$ and $F_v = -F_x \sin \psi + F_y \cos \psi$. The crosswind acts along positive y ; the headwind acts along negative x . Wind applies no yaw moment in this model. Added mass, wave excitation, and vertical motion are outside the slide model.

The original simulation source was unavailable, so the implementation is reconstructed from the equations, captions, and embedded plot images. SciPy integrates the six-state system with relative tolerance 10^{-10} , absolute tolerance 10^{-12} , and a maximum step of 0.05 s. Integration restarts at every thrust switch, and results are sampled at 0.05 s intervals. The next sections apply the same model to each slide experiment.

3. Steady and differential thrust

The first experiment isolates surge motion. Both thrusters step to 100 N at $t = 1$ s from rest. As speed rises, drag increases until it balances the 200 N total force. Setting $\dot{u} = 0$ in Equation 3 gives the terminal-speed relation

$$d_2 u_\infty^2 + d_1 u_\infty = 200, \quad u_\infty = \frac{-d_1 + \sqrt{d_1^2 + 800d_2}}{2d_2} = 1.3451 \text{ m/s.} \quad (6)$$

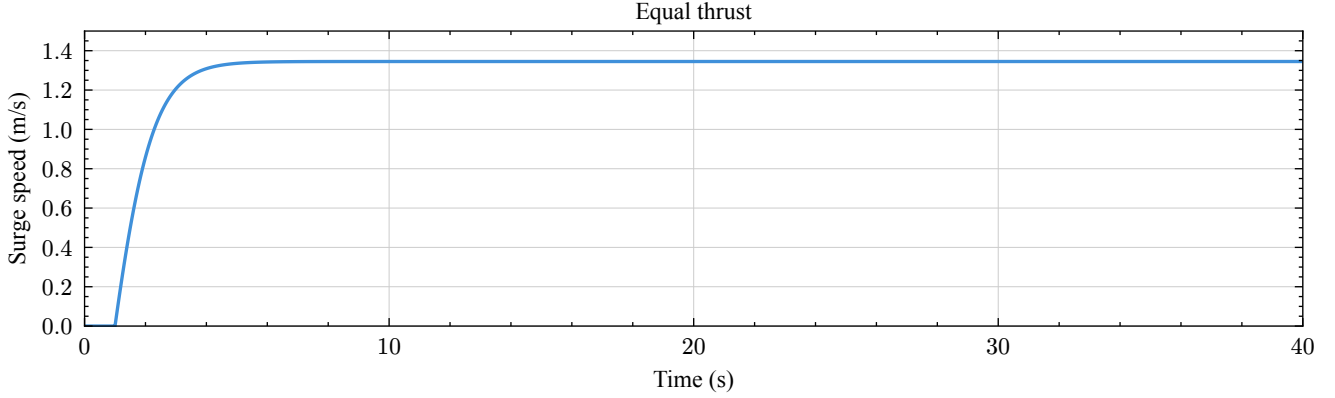


Figure 2: Equal thrust produces monotonic surge acceleration toward the drag-balanced speed. Sway and yaw remain zero.

The second experiment applies $T_L = 100$ N and $T_R = 150$ N at $t = 1$ s. The total force rises to 250 N, and the 50 N thrust difference produces a 60 N m yaw moment. Consequently, the terminal yaw rate is $r_\infty = \frac{60}{400} = 0.15$ rad/s. Rotation induces negative sway through the $-ur$ term, which feeds back into surge through vr .

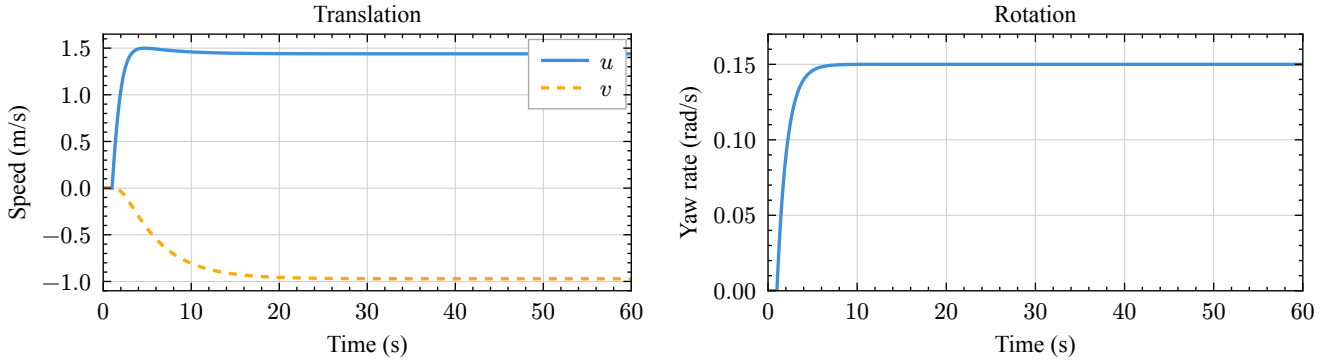
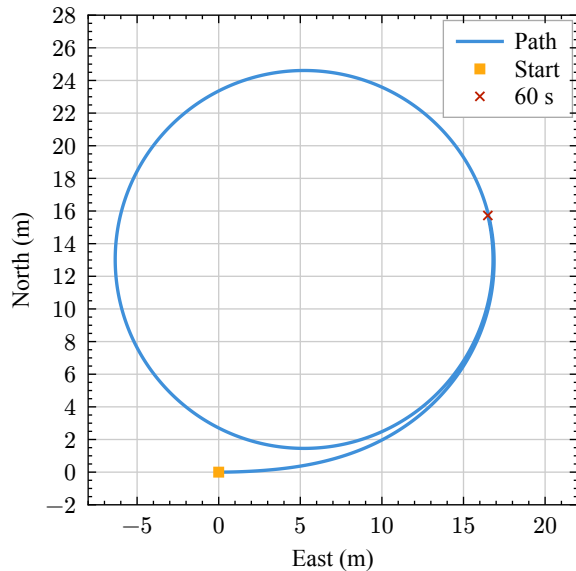


Figure 3: Differential thrust couples forward motion and turning. Translational speed and yaw rate use separate axes because their units differ.



The coupled steady speeds are $u = 1.4391$ m/s and $v = -0.9714$ m/s. From Equation 4, $v_\infty = -mu_\infty \frac{r_\infty}{d_v}$, so the lateral speed persists while the boat turns.

The corresponding path approaches a circular orbit. At 60 s, the boat is at (16.49, 15.73) m, consistent with the endpoint shown in the presentation. Equal scaling of both position axes preserves the path geometry.

The steady turning radius is approximately $\frac{\sqrt{u_\infty^2 + v_\infty^2}}{|r_\infty|} = 11.58$ m. A constant thrust difference therefore creates sustained turning rather than convergence to a target.

4. Pulse response and free decay

The steady turning response establishes the velocity coupling. A short asymmetric pulse then shows how that coupling persists after thrust is removed. The boat starts from rest, receives $T_L = 150$ N and $T_R = -100$ N over $1 \leq t < 2$ s, and coasts for the remainder of the 30 s experiment.

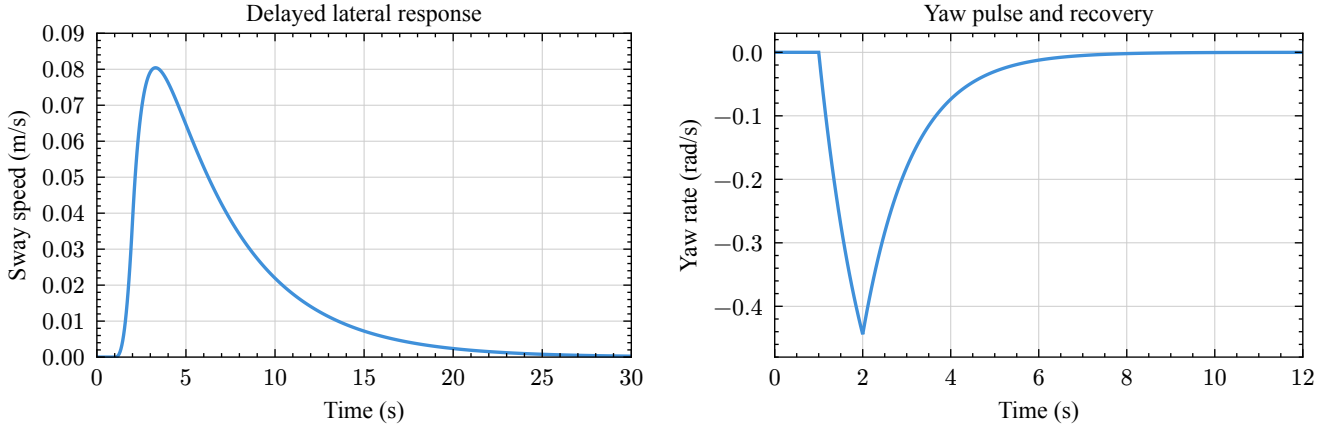


Figure 4: A one-second pulse creates a sharp yaw response and a slower sway response. The yaw panel enlarges the first 12 s to expose the transient.

The pulse applies 50 N of surge force and -300 N m of yaw moment. The reconstructed yaw minimum is -0.4441 rad/s, while sway reaches 0.0804 m/s. Positive sway follows from $-ur$ when surge is positive and yaw rate is negative. Its peak occurs after the pulse because yaw-induced translation and sway damping act on different time scales.

The unforced experiment starts with $(u, v, r) = (1, 0.5, 0.5)$ and applies zero thrust for 30 s. Each velocity approaches zero, as shown below, but the final position and heading depend on the integrated transient.

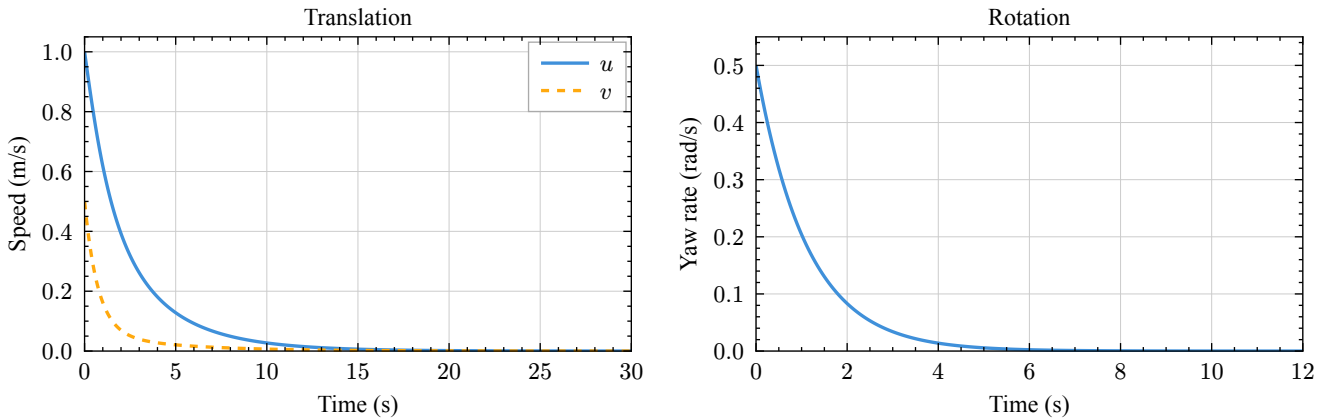


Figure 5: Free decay dissipates velocity without restoring the initial pose. Yaw decays faster than sway.

This behavior follows directly from kinetic energy. Defining $E = \frac{m(u^2 + v^2)}{2} + I_z \frac{r^2}{2}$, substitution of Equation 3, Equation 4, and Equation 5 under zero thrust and zero wind gives

$$\dot{E} = -d_1 u^2 - d_2 |u|^3 - d_v v^2 - d_r r^2 \leq 0. \quad (7)$$

The rotational coupling terms cancel because they transfer energy between surge and sway. Damping removes the remaining energy. Thus, the velocity equilibrium is attracting, while position has no restoring term. This distinction explains why the boat can stop moving without reaching the desired location.

5. Phase-plane behavior and wind drift

The decay experiment motivates a closer view of the sway–yaw subsystem. Holding surge at a fixed value U produces the reduced equations $\dot{v} = -d_v \frac{v}{m} - Ur$ and $\dot{r} = -d_r \frac{r}{I_z}$. These phase portraits use 25 initial pairs spanning $v_0 \in [-1, 1]$ m/s and $r_0 \in [-0.5, 0.5]$ rad/s. Surge is frozen for this diagnostic and is not integrated as a third dynamic variable.

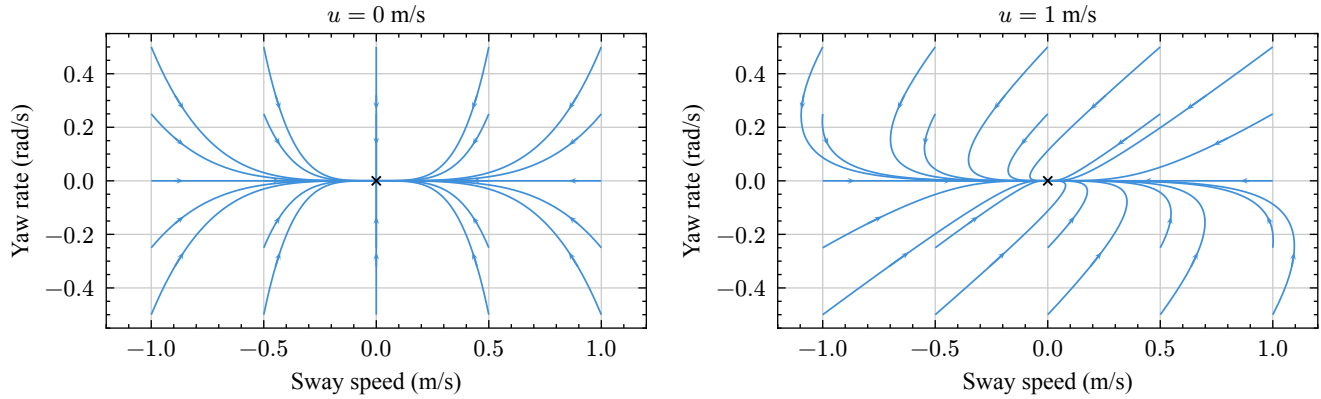


Figure 6: Sway–yaw trajectories converge to the origin for both frozen surge speeds. Nonzero surge bends the trajectories through yaw–sway coupling.

The reduced subsystem has decay rates $-\frac{d_v}{m} = -0.2222 \text{ s}^{-1}$ and $-\frac{d_r}{I_z} = -0.8969 \text{ s}^{-1}$. Both remain negative when U changes, so surge modifies the transient geometry without changing these two eigenvalues. The corresponding time constants are 4.5 s for sway and 1.115 s for yaw.

Wind changes the equilibrium rather than simply delaying decay. With no thrust, zero initial velocity, and 15.3 N of lateral wind force, Equation 4 becomes $m\dot{v} + d_v v = 15.3$. Its solution is

$$v(t) = \frac{F_y}{d_v} \left(1 - e^{-\frac{t}{\tau_v}}\right), \quad y(t) = \frac{F_y}{d_v} \left[t - \tau_v \left(1 - e^{-\frac{t}{\tau_v}}\right)\right], \quad \tau_v = \frac{m}{d_v}. \quad (8)$$

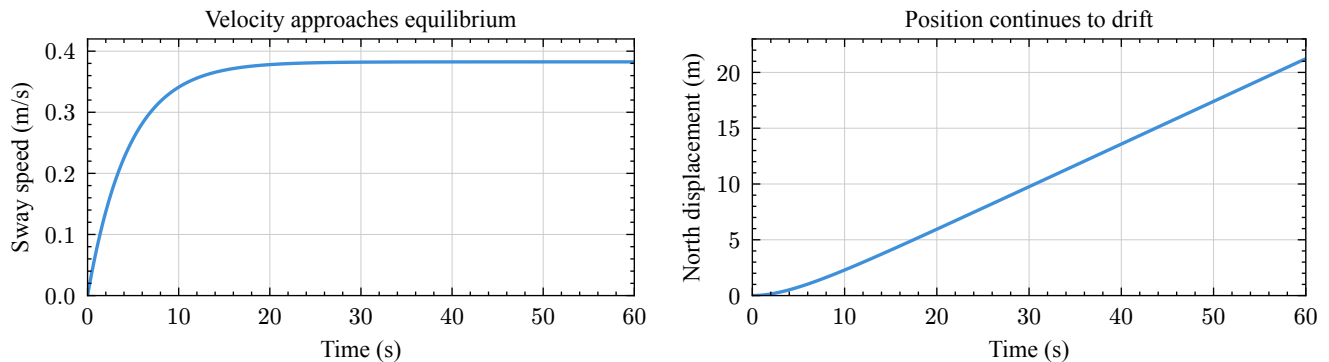


Figure 7: A constant crosswind creates a bounded lateral velocity and unbounded position drift. The same input gives both curves.

The lateral equilibrium is $v_\infty = \frac{15.3}{40} = 0.3825 \text{ m/s}$. Integrating this motion produces 21.2288 m of displacement at 60 s. Once the velocity settles, position grows almost linearly. The absence of direct lateral thrust therefore becomes a guidance problem, even when all velocity transients are damped.

6. Open-loop target approach

The wind-drift result predicts failure of a schedule calibrated only for calm water. To test that consequence, the boat starts at the origin with zero velocity and aims at (20, 0) m. Both thrusters apply 100 N for 13.8329 s and then switch to zero. This schedule is reconstructed from the calm slide trace because the original switching schedule was not stated. The 200 N force is supported by a trace fit of 200.58 N; the cutoff is selected so that the thrust interval plus the subsequent coast covers 20 m.

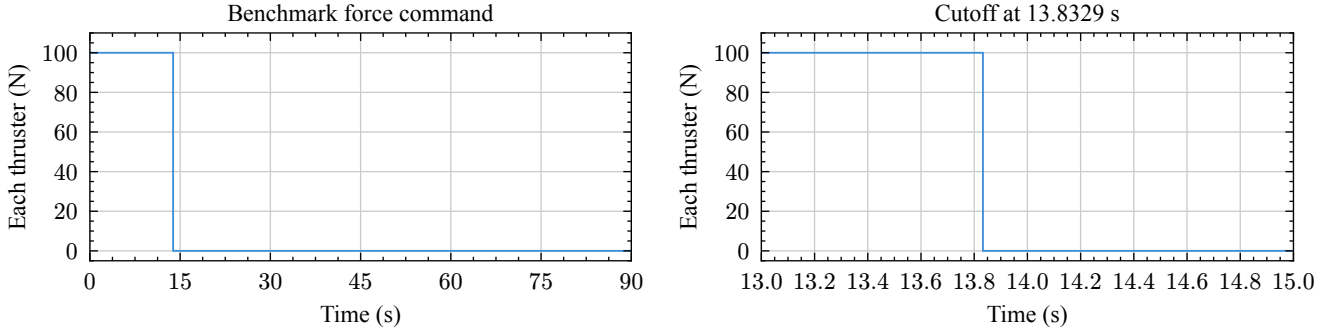


Figure 8: Expanded benchmark schedule and cutoff detail. Both force commands are equal, so the allocated yaw moment is zero. Thrust is instantaneous in these original experiments; the later component tests add actuator lag.

After the switch, the calm-water coast distance from speed u_s is $\frac{m}{d_2} \ln\left(1 + d_2 \frac{u_s}{d_1}\right)$. Adding it to the distance accumulated during thrust gives the selected 20 m endpoint. The wind cases then use the same schedule without further fitting.

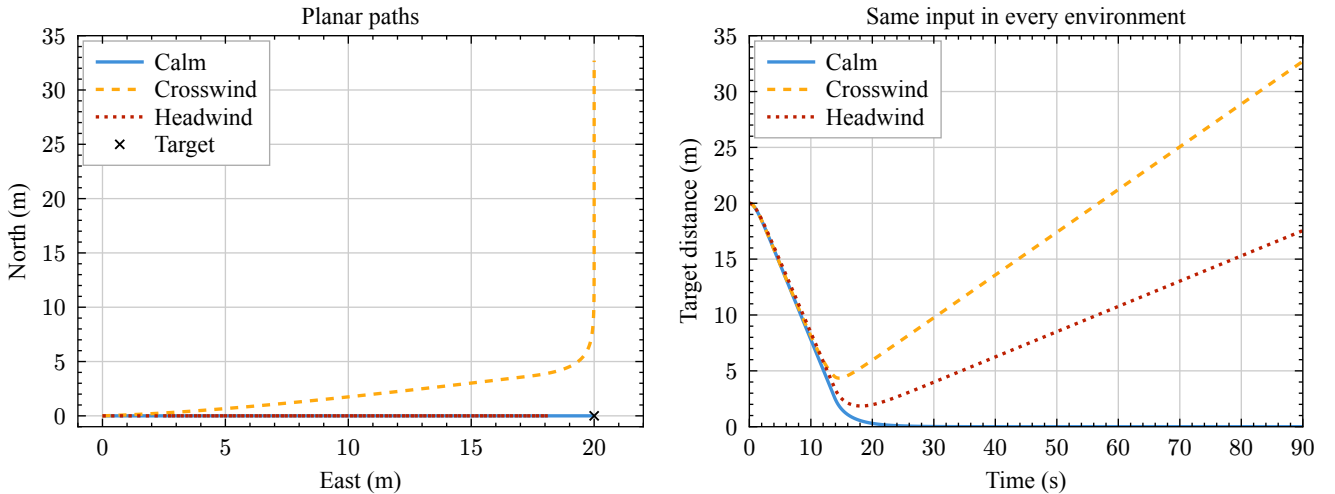


Figure 9: The nominal schedule reaches the calm-water target but accumulates large errors under constant wind. The path panel uses separate axis scales; the distance panel gives the physical error.

Environment	$x(90)$ (m)	$y(90)$ (m)	Target error (m)
Calm	20.0000	0.0000	0.0000
Crosswind	20.0000	32.7038	32.7038
Headwind	2.4482	0.0000	17.5518

Crosswind preserves the nominal forward motion because heading remains zero, but lateral displacement reaches 32.70 m. Headwind reduces progress during thrust and drives the boat backward after thrust ends. The final headwind surge speed is -0.2261 m/s. Both cases show why a preplanned input alone cannot regulate target distance in a changing environment.

7. Reproduction and numerical verification

The preceding experiments retain the slide parameters and reproduce their visible trajectories. To quantify agreement, the embedded images are extracted unchanged from the presentation, and colored traces are digitized using recorded axis calibrations. Legend regions are excluded. The root-mean-square (RMS) differences below compare reconstructed curves with the retained raster samples; the largest is 0.81 pixel.

Slide curve	Samples	RMS in physical units	RMS pixels
Equal thrust, u	414	0.0009 m/s	0.281
Differential thrust, u	622	0.00254 m/s	0.506
Differential thrust, v	620	0.00128 m/s	0.256
Differential thrust, r	397	0.0007 rad/s	0.14
Pulse, v	762	0.00009 m/s	0.346
Pulse, r	207	0.00059 rad/s	0.421
Free decay, u	402	0.00145 m/s	0.609
Free decay, v	338	0.00191 m/s	0.806
Free decay, r	190	0.00158 rad/s	0.666
Crosswind drift, y	772	0.01253 m	0.249
Target distance, calm	140	0.0293 m	0.44
Target distance, crosswind	494	0.02824 m	0.425
Target distance, headwind	562	0.01683 m	0.253

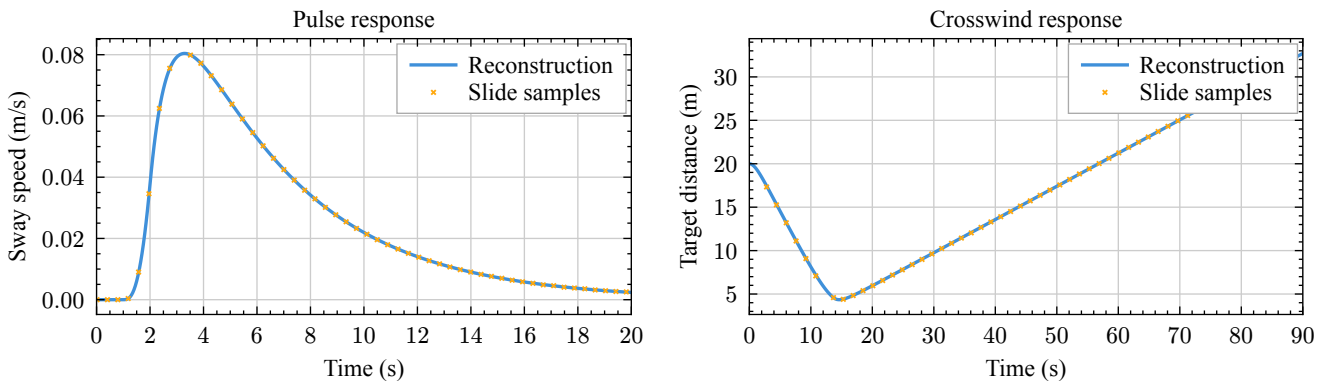


Figure 10: Representative overlays show the rebuilt curves against sampled slide traces. The comparison uses every retained sample; only every tenth marker is displayed.

Independent checks support the numerical implementation. The surge step agrees with its exact scalar solution to 3.18×10^{-11} m/s, the yaw step agrees to 3.46×10^{-13} rad/s, and the wind-drift solution agrees to 2.40×10^{-14} m. Halving the maximum integration step changes no state component by more than 2.38×10^{-10} in its corresponding units. The free-decay energy decreases monotonically, and every thrust input remains within the stated limits.

The accompanying project stores the simulation code, input schedule, numerical checks, digitized samples, source-image hashes, and figure source. Consequently, the report can be rebuilt from the retained presentation rather than depending on inaccessible plotting code. Yaw damping and the blind schedule remain explicitly identified as reconstructed quantities.

8. Selected hardware and mission architecture

The benchmark shows why feedback is required. Real-time kinematic (RTK) positioning supplies the global navigation satellite system (GNSS) reference. The deployment architecture assigns image processing and guidance to AGX Orin while retaining the BlueBoat Pi 4, Navigator, and ArduRover for low-level control. This assignment keeps the motor loops independent of the image-processing workload [2].

Function	Selection	Reason / interface
Companion	AGX Orin, 64 GB configuration [4]	Perception, fusion, guidance; Ethernet
Stereo vision	One ZED X, 2 mm wide lens	Global shutter; GMSL2 / ZED Link Duo capture
Inertial sensing	VectorNav VN-110 rugged	Gyroscope, acceleration, attitude; RS-422
Position / heading	Septentrio mosaic-H	RTK position; dual-antenna heading
Antennas	Two PolaNt-xMF [5]	Fore-aft mounting, ≈ 1 m baseline
Propulsion	Existing M200 pair and ESCs	Navigator pulse-width modulation

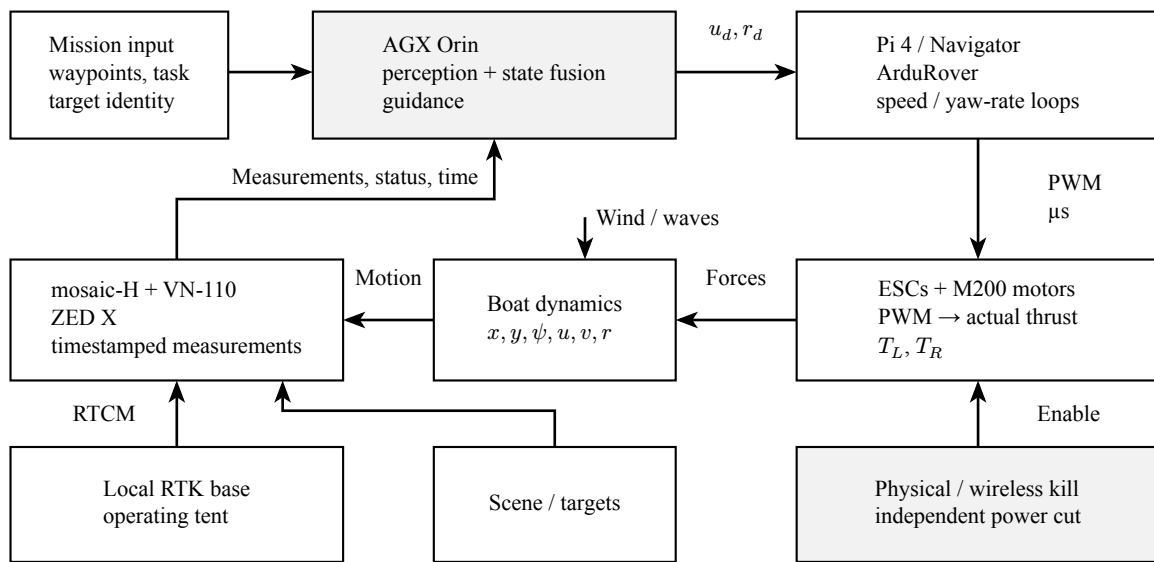


Figure 11: High-level deployment architecture. Mission information enters guidance; sensors return measured motion and scene information. Guidance sends speed and yaw-rate references to the autopilot. The power-cut path acts independently of the companion computer.

- One ZED X is the initial camera configuration [6].
 - It already contains two synchronized imagers and a 120 mm stereo baseline.
 - A second ZED X adds rear or side coverage. It is justified when a measured blind area limits a task, rather than to obtain the first stereo depth estimate.
 - Use the ZED Link Duo capture card supported on AGX Orin [7]. Reserve a second mounting point and validate synchronization, extrinsics, and execution load before enabling it.
- RTK uses a local correction source [8].
 - Mount a reference receiver at a surveyed point in the operating tent and relay Radio Technical Commission for Maritime Services (RTCM) corrections over the permitted local link.
 - Dual antennas give heading while stationary; the receiver reports solution status and correction age with the position.

9. Signal ownership and measured outputs

The high-level blocks in Figure 11 separate responsibility. The detailed interfaces below make each command and measurement explicit, so guidance inputs are not confused with motor inputs.

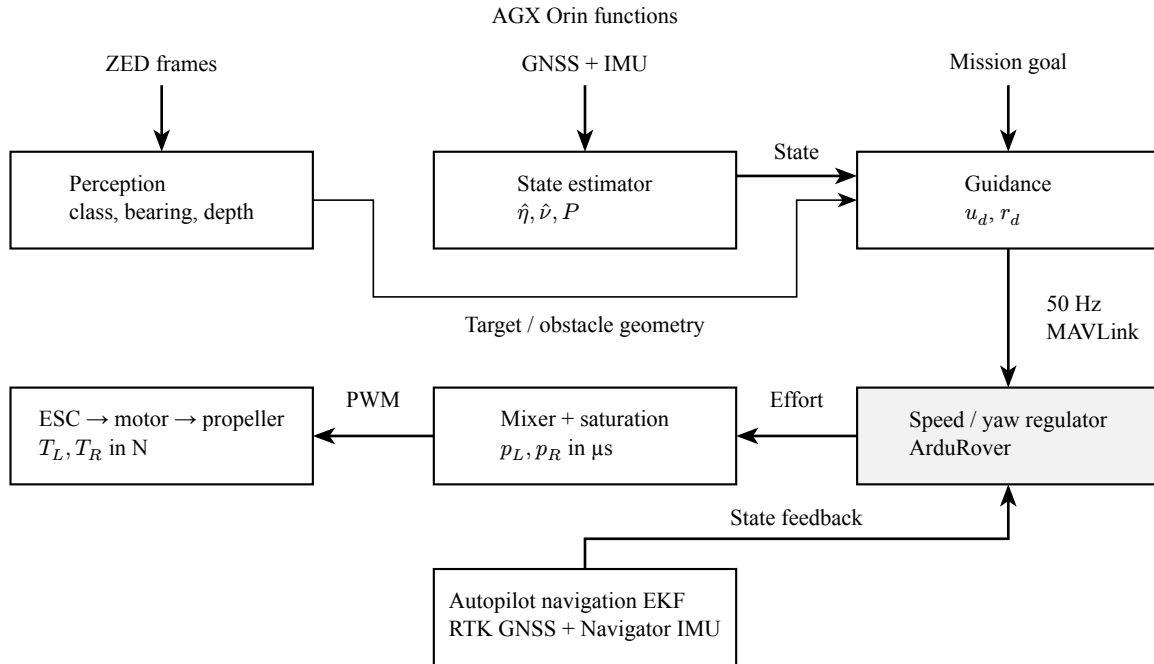


Figure 12: Two control levels. Orin uses sensor observations and the mission goal to produce feasible u_d and r_d . ArduRover closes its own speed and yaw-rate loops, mixes the effort, and sends motor pulse widths. Perception supplies scene geometry directly to guidance.

Boundary	Input	Output
Mission → autonomy	Waypoints, target class, task state	Selected target / route
Perception	Stereo images, calibration, timestamps	Class, bearing, depth, validity
State estimator	GNSS, gyro, acceleration, attitude	$\hat{x}, \hat{y}, \hat{\psi}, \hat{u}, \hat{v}, \hat{r}$ and covariance
Guidance	Route, target geometry, estimated state	u_d in m/s; r_d in rad/s
Autopilot	References; its navigation feedback	Left / right PWM in μs
Physical plant	Actual T_L, T_R ; wind / waves	Pose and body velocity

- The physical plant is actuated by force, not by sensor data.
 - An electronic speed controller (ESC) turns a pulse-width modulation (PWM) command into motor drive. The propeller converts rotation into thrust.
 - Actual thrust is a modeled internal quantity, inferred from the installed map. This sensor suite does not directly measure thrust, torque, or motor revolutions per minute.
- The navigation outputs are observations of boat motion.
 - GNSS measures antenna position, inertial velocity, and dual-antenna heading. Orin and the autopilot both consume the receiver stream; Navigator's IMU supplies autopilot propagation. The VN-110 supplies Orin's independent motion observations.
 - The inertial measurement unit (IMU) measures angular velocity and specific force. Its attitude and heading reference system (AHRS) provides an orientation estimate.
 - Use dual-GNSS heading as the main absolute heading reference. Motor magnetic fields can contaminate magnetic heading.
- The camera measures the scene relative to its optical frame.
 - Images are the raw outputs; detection class, bearing, disparity, and depth are processed outputs. Global position requires the estimated boat pose and camera extrinsics.

10. Navigation sensing and reference conventions

The measured outputs become useful to control only after their coordinate frames and timestamps agree. Configure the autopilot's Septentrio Binary Format dual-antenna backend and GNSS yaw source, then verify received rates and antenna offsets [9]. Mount the VN-110 close to the center of gravity, survey both antenna lever arms, and calibrate the camera-to-hull transform. Use the GNSS pulse-per-second signal to discipline timestamps and record acquisition and delivery times separately.

- The mosaic-H supplies the high-accuracy global reference [8].
 - The manufacturer specifies horizontal RTK accuracy of 0.6 cm plus 0.5 parts per million of baseline distance and heading accuracy of 0.15° with 1 m antenna separation.
 - The component test assumes RTK fixed, 20 Hz updates, 2 cm position noise, and 80 ms delivery latency. It uses 0.03 m/s velocity noise and 0.15° heading noise.
 - Inflate the estimator covariance when RTK becomes float or single-point. Stop using a stale correction status as evidence of centimeter positioning.
- The VN-110 supplies rapid motion measurements [10].
 - Configure 200 Hz raw IMU output. The simulation uses 5 ms delivery latency, 0.002 rad/s gyro noise, a 0.001 rad/s residual bias, and 0.02 m/s^2 acceleration noise.
 - These are integration stress assumptions. They are not derived from the manufacturer's noise-density specification.
 - Magnetic AHRS heading is simulated at 2° standard deviation, matching the scale of the manufacturer's 2° RMS heading specification under calibrated magnetic conditions.

For a primary antenna located at body offset $\ell = (-0.5, 0)^T \text{ m}$, the observation model accounts for both position and rotational velocity.

$$\mathbf{p}_a = \mathbf{p} + R(\psi)\ell, \quad \mathbf{V}_a = R(\psi)[\mathbf{v}_b + rJ\ell], \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (9)$$

Here $\mathbf{p} = (x, y)^T$, $\mathbf{v}_b = (u, v)^T$, and $R(\psi)$ is the planar rotation matrix. A gyro measures $r + b_g + n_g$. At the center of gravity, the planar accelerometer model after gravity compensation is

$$a_x = \dot{u} - vr + n_x, \quad a_y = \dot{v} + ur + n_y. \quad (10)$$

This rotating-frame correction avoids treating a body velocity derivative as the accelerometer output. Roll, pitch, gravity, and off-center acceleration terms must be restored for the three-dimensional deployment.

The report uses east–north coordinates with leftward sway and counterclockwise yaw. ArduRover accepts body-forward velocity and yaw rate through the MAVLink message SET_POSITION_TARGET_LOCAL_NED in Guided mode [11]. Use the velocity-plus-yaw-rate mask 1511 and body frame 9. For this convention, set $v_x = u_d$, $v_y = 0$, and $\text{yaw_rate} = -r_d$. If sending absolute heading, convert with $\psi_{\text{NED}} = \frac{\pi}{2} - \psi$. The adapter sends the 50 Hz references and verifies the sign conversion in a low-speed left-turn test.

11. Computer execution and response budget

The command interface defines what must meet a deadline. Run state propagation and guidance at 50 Hz, process images at 30 Hz, and deliver the latest valid reference through a bounded queue. Camera processing uses a separate worker so a late image cannot hold the guidance loop.

Budget, followed by thrust lag $\tau_T = 0.2$ s

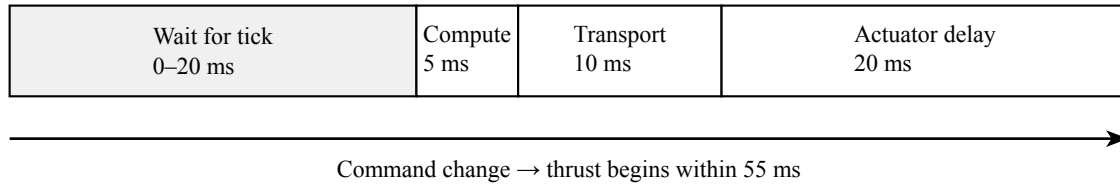


Figure 13: Command-to-thrust-onset timing budget. Widths represent the allocated time intervals. The motor then approaches its static force with a separate first-order lag. All latency values are design assumptions.

Stage	Budget	Meaning
Guidance sampling	0–20 ms	Wait until the next 50 Hz tick
Estimator / guidance compute	5 ms	Execution allowance per tick
Command transport / acceptance	10 ms	Orin to low-level controller
Actuator dead time	20 ms	PWM update and drive response
Motor time constant, τ_T	0.2 s	Sensitivity tests at 0.1 and 0.4 s

For a newly changed command, the onset delay is bounded by the sample wait plus the fixed delays. The nominal bound is

$$L_{\text{onset}} \leq 20 + 5 + 10 + 20 = 55 \text{ ms.} \quad (11)$$

A first-order motor reaches 90% of its final force after $\tau_T \ln 10 = 0.4605$ s from onset. The simulated command changes at 1.023 s, is sampled at 1.040 s, and starts changing thrust at 1.075 s. The sampled 90% crossing occurs 0.517 s after demand, including motor lag.

- AGX Orin performance is established by execution measurements.
 - The 5 ms guidance budget and 20 ms vision budget are targets. Processor throughput does not establish either latency.
 - Log acquisition, dequeue, compute start/end, command send/receive, and PWM application timestamps using a common clock.
 - Record median, 95th and 99th percentile, maximum latency, missed deadlines, power mode, temperature, and image resolution under sustained load.
- Freshness is part of the response model.
 - The assumed worst measurement ages are 130 ms for GNSS, 10 ms for IMU, and 63.3 ms for the camera processing stream.
 - These bounds add one sample period to delivery latency. Propagate the state to command time and reject stale perception updates.
 - Implement a reference watchdog with an initial 100 ms timeout. Validate neutral-command delivery and the independent kill path; ArduRover’s documented velocity-command timeout is 3 s [11].

12. Motor and propeller response

The execution budget delays a command before the propulsion system responds. The propulsion model then maps each pulse width p_i to a static thrust $T_{\{s,i\}}$ and applies a dynamic lag. The manufacturer's M200 weedless-propeller curve at 16 V supplies the signed static lookup [12], [13].

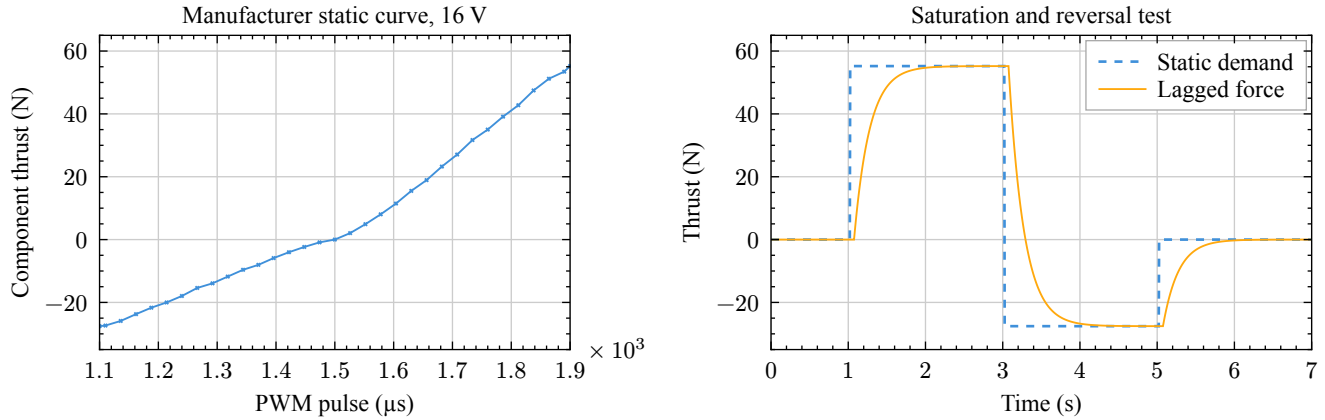


Figure 14: M200 component response. Left shows manufacturer static samples converted from kilogram-force to newtons. Right applies an over-limit forward command, an over-limit reverse command, and neutral; saturation, delay, and assumed lag determine the force response.

The static map interpolates the retained data after clipping p_i to 1100–1900 μs . Commands from 1475 through 1525 μs produce zero demand, with 1500 μs neutral. The sampled component limits at 16 V are -27.56 N reverse and 55.21 N forward. The dynamic model is

$$\tau_T \dot{T}_i + T_i = T_{\{s,i\}}(p_{i(t-L_i)}, 16\text{V}), \quad i \in \{L, R\}. \quad (12)$$

For a constant demand after onset t_0 , the force response is $T_{i(t)} = T_s + (T_{i(t_0)} - T_s)e^{-\frac{t-t_0}{\tau_T}}$. The simulation changes the pulse demand at 1.023, 3.023, and 5.023 s; it integrates exactly between delayed changes. Its first forward step agrees with this analytic response to $1.81 \times 10^{-9}\text{ N}$.

- Component thrust and installed-boat thrust require separate calibration.
 - The curve describes the manufacturer's M200-and-propeller component test. BlueBoat's published total static thrust is 8.2 kgf, approximately 80.4 N [14].
 - The two component maxima sum to 110.4 N, approximately 37% above the published boat rating. The test conditions differ; the installed map must be measured.
 - Mounting, guards, hull interaction, voltage, and inflow change the installed map. Use a load cell to measure each side and both sides together with the final guards installed.
- The dynamic model has identifiable limits.
 - Fit separate acceleration, reversal, and coast constants from synchronized force and PWM logs; test several pulse amplitudes and battery voltages.
 - Add rate limits or a second-order rotor model when the measured response requires them. The 0.2 s lag is a starting assumption, not a manufacturer response specification.

13. Boat response and parameter identification

The motor tests establish a force history. Applying that history to the six-state slide plant exposes the consequence of finite actuator response without changing the recovered benchmark. Equal thrust uses 20 N per side; the turning test uses 20 N left and 30 N right, followed by neutral.

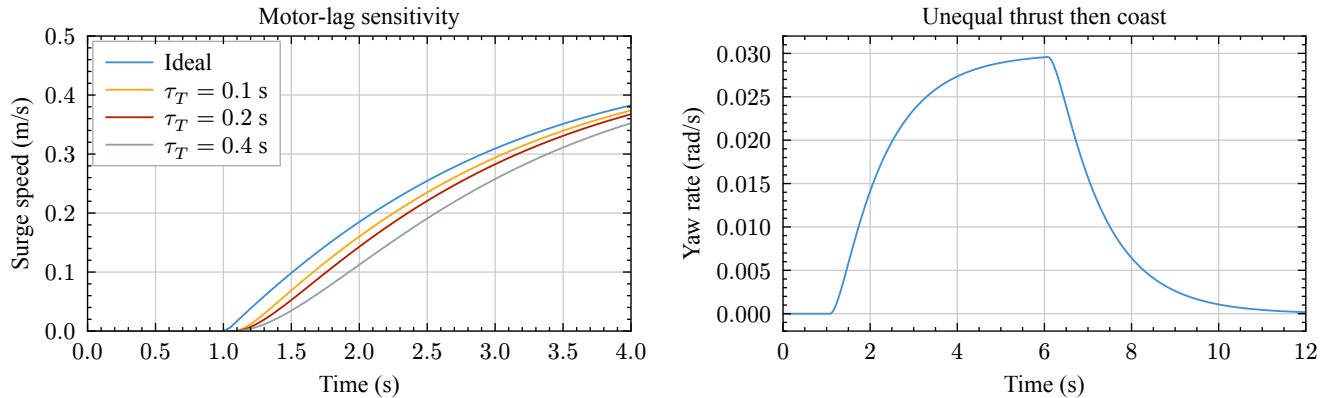


Figure 15: Component-chain tests on the slide hydrodynamic plant. Motor lag delays surge acceleration and yaw buildup. These histories compare integration assumptions; the 180 kg benchmark plant does not predict BlueBoat motion.

At 2 s, ideal equal thrust gives 0.1850 m/s while the delayed $\tau_T = 0.2$ s chain gives 0.1429 m/s. Thus, an instantaneous-thrust simulation overstates early progress by 0.0421 m/s for this input. The yaw transient also continues after neutral while the propeller force decays.

- The BlueBoat model uses measured physical parameters.
 - Weigh the boat with batteries, guards, computer, and sensor mounts. The manufacturer allows 15 kg for batteries plus payload above the 14.5 kg bare hull [2].
 - Measure the force-line spacing B and loaded center of gravity; identify yaw inertia and added mass.
 - Overall beam is not the thrust moment arm. A narrowed competition hull requires a new measurement.
- Identify the planar force response from controlled tests.
 - Equal-pulse steps establish surge acceleration and the linear/quadratic drag terms. Coast-down isolates drag from propeller force.
 - Opposing and unequal thrust establish yaw response; turning runs identify sway coupling and lateral damping.
 - Repeat in both directions and record voltage, wind, current, payload, RTK status, and temperature. Hold out complete runs for prediction checks.
- Fit a deployment model that retains hydrodynamic inertia.
 - Use $(M_{RB} + M_A)\dot{\nu} + C(\nu)\nu + D(\nu)\nu = \tau + \tau_d$ with the same planar kinematics [3].
 - Estimate surge, sway, and yaw drag with installed thrust maps; validate trajectories and transient error against RTK/IMU measurements.
 - Use a three-dimensional model when roll, pitch, wave motion, or camera stabilization materially changes sensing or actuation.

Because these parameters have not been measured on the selected hull, the report does not assign the slide mass, inertia, drag coefficients, or wind force to BlueBoat. The component functions and exported input data provide the calibration structure; measured logs supply its deployment coefficients.

14. Sensor response under changing motion

The force chain produces motion that each sensor samples at its own rate. The following test uses the same unequal-thrust maneuver as the preceding section, a fixed random seed, and acquisition-time truth. Measurements are plotted at delivery time, exposing delay rather than hiding it with an ideal continuous observation.

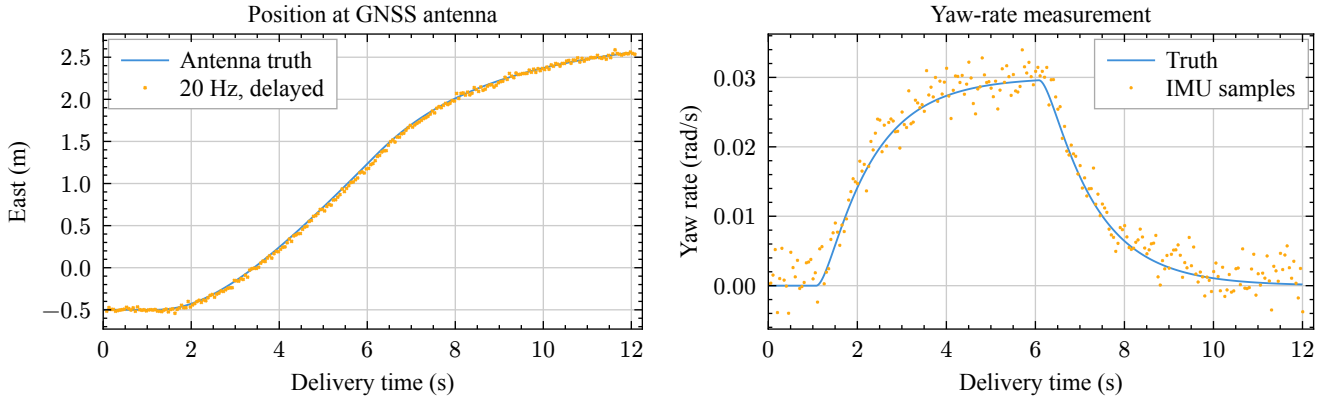


Figure 16: GNSS and IMU observation tests. Position is measured at the antenna 0.5 m aft of the hull center. The IMU updates at 200 Hz; every tenth sample is displayed. Noise and latency are the stated integration assumptions.

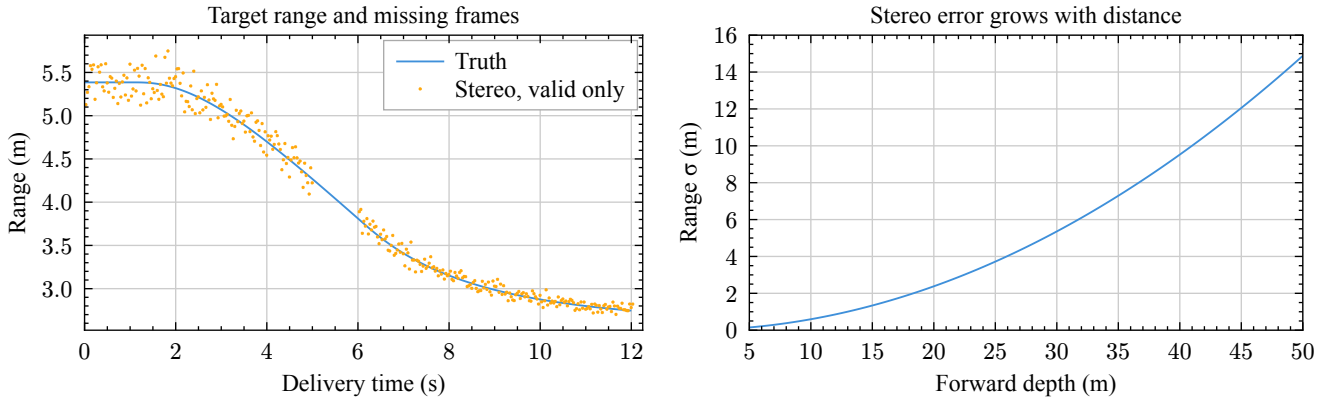


Figure 17: Stereo observation and range sensitivity. The target is at (5, 2) m in the benchmark frame; frames from 5 to 6 s are deliberately occluded. Range samples are returned only when valid. The error curve uses $f = 700$ pixels, $b = 0.12$ m, and disparity noise $\sigma_d = 0.5$ pixel.

For a rectified stereo pair, disparity $d = p_L - p_R$ gives forward optical depth $Z = f \frac{b}{d}$. A target at planar bearing β has horizontal image coordinate $p_L = c_x - f \tan \beta$ and range $\rho = \frac{Z}{\cos \beta}$. The first-order depth uncertainty is

$$\sigma_Z \approx \frac{Z^2}{fb} \sigma_d. \quad (13)$$

It is 0.60 m at 10 m depth and 2.38 m at 20 m under this disparity-noise assumption. The camera model therefore reports range validity rather than promising centimeter accuracy from stereo.

- Camera latency includes capture and perception.
 - The test processes 30 frames/s with 10 ms acquisition/transfer and 20 ms vision compute. It assumes a rectified 1920-pixel image with $f = 700$ pixels; deployed intrinsics replace that approximation.
 - Invalid, behind-camera, out-of-view, occluded, and beyond-20 m observations are withheld. The run returns 331 of 361 frames.
 - Water glare, low texture, and object occlusion require measured validity thresholds. This geometric test does not reproduce the detector or the stereo software's full error distribution.

15. Control feasibility and competition integration

The expanded component chain supports controller development, but it does not supply closed-loop performance results. The presentation's proposed linear-quadratic regulator (LQR) minimizes state error and control effort around a local operating point [15]. Its feasibility depends on whether thrust can influence each relevant motion mode.

- The boat is laterally underactuated.
 - At rest, $\dot{y} = v$ and $\dot{v} = -d_v \frac{v}{m}$ have no direct thrust input. The six-state controllability matrix has rank four; the uncontrolled lateral-position integrator has eigenvalue zero.
 - At a straight-running reference of $U = 1$ m/s, yaw and lateral motion couple and the rank becomes six.
 - Use a moving reference and constrained guidance. A full-state LQR about rest cannot stabilize arbitrary lateral position.
- The initial deployment closes the motor loops in ArduRover.
 - Orin sends feasible speed and turn-rate references. Route tracking or an outer-loop regulator can be evaluated without introducing a competing motor controller.
 - An LQR that outputs individual forces requires a separate low-level implementation, allocation, saturation handling, and validated failsafes.
 - The slide targets of under 30 s recovery, under 10% overshoot, under 0.3 m position error, and under 15° heading error remain proposed targets, not achieved results.

The competition configuration also changes the boat's physical model. The 2027 event is listed for February 18–23 in Sarasota, Florida; its handbook is pending as of October 8, 2026 [16]. The available 2026 requirements define the present integration checks [17], [18].

- Geometry must fit the competition envelope.
 - The published BlueBoat beam is 0.93 m; the 2026 width limit is 3 ft, or 0.9144 m. The nominal hull exceeds that limit by 15.6 mm.
 - Measure the assembled hull and guards. Obtain the current rule interpretation or design narrower crossbars before claiming eligibility.
- Propulsion must include guards and independent stopping.
 - The requirements specify protected propellers and physical and wireless kill functions that disconnect motor/actuator power.
 - Add guards to the final mounting layout, then remeasure thrust. Route the kill circuit outside the Orin software path shown in Figure 11.
- Power and navigation require installation-level checks.
 - Use regulated, fused power and thermal management for Orin and the GMSL2 capture hardware; the hull's 5 V auxiliary rail is rated at 5 A [2].
 - Use local RTK corrections from equipment in the operating tent; the 2026 rules prohibit outside internet connections, including LTE corrections, during semifinal and final runs.
 - Recheck these requirements against the 2027 handbook when released.

These checks lead directly to the next experiment. Install the selected sensor stack, synchronize its logs, identify the loaded hull and guarded propulsion response, and benchmark the execution deadlines. Then evaluate feasible guidance and feedback against the original recovery targets with measured component parameters.

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